

# **WG21 2021-02 Virtual Meeting**

## **Record of Discussion**

ISO/IEC JTC1 SC22 WG21 P2330 — 2021-03-08

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Chair: John Spicer  
SC22/WG21 pre-meeting telecon

Monday 2021-02-22  
08:00 N.Am. Pacific Time

### **1. Opening activities**

John Spicer opens the meeting at 08:03 AM N.Am. Pacific Time

#### **1.1 Opening comments (PL22.16)**

John Spicer presents.

Thank you to everyone for your continuing efforts in these challenging times.

#### **1.2 Meeting guidelines**

John Spicer presents.

Meetings are not public, but are open to visitors. Please refrain from live tweeting, blogging, taking photos or videos.

Every participant is responsible for understanding and abiding by the following:

- [The INCITS Antitrust Guidelines \(PL22.16\)](#)
- [The INCITS Patent Policy \(PL22.16\)](#)
- [The ISO Code of Conduct](#)
- [The INCITS Code of Conduct \(PL22.16\)](#)
- [The IEC Code of Conduct](#)
- [The WG21 Practices and Procedures, and Code of Conduct](#)

John Spicer presents the meeting guidelines. We expect everyone to be familiar with these rules. These apply to the plenary and to all the subgroup meetings

If you have any questions or concerns about CoC issues, please approach a committee officer or a NB representative and bring it to their attention. If you have any technical issues or concerns, please bring them up as soon as possible.

### **1.3 The ISO Code of Conduct**

ISO requires that, through 2020, committees provide an opportunity to discuss the code of conduct.

John Spicer presents ISO CoC slides.

Please report any CoC violation to @conduct even if you also report it to the subgroup chair.

### **1.4 Membership, voting rights, and procedures for the meeting (PL22.16)**

John Spicer presents voting rights.

If you are representing an organization that is considering formally joining PL22.16, or your organization is already a member and you wish to change your voting status, please inform an officer.

Hal Finkel presents. When registering for the call, please make sure you use the name that is registered in the global directory.

John Spicer presents how to vote using the telecon client.

### **1.5 Introductions**

Officers introduce themselves.

New members introduce themselves.

John Spicer welcomes new members.

Herb Sutter: We have 18 national bodies present at this virtual meeting.

### **1.6 Agenda review and approval (PL22.16 motion, WG21 poll)**

The meeting goals described above are derived from the schedule adopted in 2020 and described in: [P1000R4](#)

The primary goal of this meeting will be to provide any necessary status updates and conduct straw polls proposed for working draft changes.

John Spicer presents the agenda.

PL22/16 motion to approve the meeting agenda  
Barry Headquist Moves  
Jonathan Wakely seconds.  
The motion is unanimously approved by PL22/16.

WG21 motion to approve the meeting agenda, with the changes to the agenda as discussed.  
The motion is unanimously approved by WG21.

### 1.7 Editor's reports, approval of working drafts

| Document                | Editor's report       | Prospective WD        |
|-------------------------|-----------------------|-----------------------|
| C++23 Standard          | <a href="#">N4879</a> | <a href="#">N4878</a> |
| Library Fundamentals TS | <a href="#">N4874</a> | <a href="#">N4873</a> |

WG21 motion to approve the working drafts.  
The motion is unanimously approved by WG21.

### 1.8 Approval of the minutes of the previous meetings (PL22.16 motion, WG21 poll)

| Meeting                                          | Minutes                            |
|--------------------------------------------------|------------------------------------|
| WG21 November Virtual                            | <a href="#">N4877</a>              |
| PL22.16 November Virtual                         | <a href="#">pl22.16-2020-00009</a> |
| WG21 pre-February Virtual administrative telecon | <a href="#">N4883</a>              |

PL22/16 motion to approve the minutes.  
Daveed Vandevoorde moves.  
Nevin Liber seconds  
The motion is unanimously approved by PL22/16.

WG21 motion to approve the minutes.  
The motion is unanimously approved by WG21.

2. Liaison reports, and WG21 study group reports (see pre-meeting WG21 telecon minutes)

No discussion.

3. WG progress reports (Core, Evolution, Library, Library Evolution; see pre-meeting WG21 telecon minutes)

No discussion.

#### 4. New business requiring action by the committee

No discussion.

### 5. Discussion and Straw Polls

#### 5.1 CWG Polls

Mike Miller presents.

Regarding CWG motion 1 : we adopted P2238 at the last plenary, but at that point we have not considered whether each individual fix should be a DR against C++20 or not. We have considered it since and have decided all the resolved issues should be considered as DRs against C++20.

We only had 2 meetings since the last plenary so we do not have many other issues resolved at this point in time.

**1. Accept as Defect Reports the the following issues (all issues resolved by [P1787R6](#) (Declarations and where to find them), adopted at the November, 2020 meeting):**

|      |      |      |      |      |      |      |      |      |      |
|------|------|------|------|------|------|------|------|------|------|
| 36   | 110  | 138  | 191  | 255  | 271  | 279  | 338  | 360  | 386  |
| 399  | 405  | 418  | 536  | 554  | 562  | 563  | 600  | 607  | 852  |
| 952  | 1028 | 1200 | 1252 | 1291 | 1478 | 1500 | 1616 | 1729 | 1771 |
| 1818 | 1820 | 1821 | 1822 | 1828 | 1829 | 1835 | 1837 | 1839 | 1841 |
| 1884 | 1894 | 1896 | 1898 | 1900 | 1907 | 1908 | 1936 | 2007 | 2009 |
| 2058 | 2062 | 2065 | 2070 | 2165 | 2199 | 2213 | 2331 | 2370 | 2396 |
| 2413 |      |      |      |      |      |      |      |      |      |

No discussion.

No objection to unanimous consent.

Motion passes.

**2. Accept as Defect Reports all issues in [P2313R0](#) (Core Language Working Group "tentatively ready" Issues for the February, 2021 meeting) and apply the proposed resolutions to the C++ working paper.**

No discussion.

No objection to unanimous consent.

Motion passes.

**3. Apply the changes in [P1102R2](#) (Down with (!)) to the C++ working paper.**

No discussion.

No objection to unanimous consent.

Motion passes.

## 5.2 LWG Polls

Jonathan Wakely presents.

I would like to call out motion 4 which has a new version attached to the wiki.

John Spicer : Do we need further clarification ?

Nico Jossutis : Is this a semantic or syncatic change ?

Barry Revzin : there are no semantic changes, we are saying the same thing better.

Jonathan Wakely : LWG reviewed the paper, asked for some changes and approved those changes, but it didn't get into the mailing. Barry spotted this last week and corrected it. The correct version was uploaded to the wiki, but the motion wasn't updated until today. In terms of semantic changes, there are none.

John Spicer : the policy is to explain any changes that happen to the straw polls after the deadline, regardless of the magnitude of changes.

### *Issues*

**1. Apply the changes for all Tentatively Ready issues in [P2315R0](#) (C++ Standard Library Issues to be moved in Virtual Plenary, Feb. 2021) to the C++ working paper.**

No discussion.

No objection to unanimous consent.

Motion passes.

## **C++23**

**2. Apply the changes in [P2259R1](#) (Repairing input range adaptors and counted\_iterator) to the C++ working paper.**

No discussion.

No objection to unanimous consent.

Motion passes.

**3. Apply the changes in [P2212R2](#) (Relax Requirements for time\_point::clock) to the C++ working paper.**

No discussion.

No objection to unanimous consent.

Motion passes.

**4. Apply the changes in [P2162R2](#) (Inheriting from std::variant) to the C++ working paper.**

Peter Brett : I disagree that this fixes the problem.

John Spicer : has this been raised as a concern?

Peter Brett : I have mentioned it to some people

Herb Sutter : Bryce, was this question considered ?

Bryce Adelstein Lelbach : I do not know if have discussed this. I will check.

There are objections in the room.

Nico Josuttis : to be able to vote, I need to know if there is a problem if we were to postpone this paper.

Jonathan Wakely : we haven't discovered anything new. The objection is that the rules have not been relaxed enough.

Bryce Adelstein Lelbach : I just reviewed the minutes. Yes, this was discussed in LEWG. We did discuss future extensions, and we believe that nothing we do now precludes future extensions.

Ville Voutilainen : if people want these future extension, they should vote yes as this change helps those future extensions.

Tony van Eerd : I have also brought this up in an email.

John Spicer : given the discussion, are there still objections in the room

No objection to unanimous consent.

Motion passes.

**5. Apply the changes in [P2160R1](#) (Locks lock lockables) to the C++ working paper.**

No discussion.

No objection to unanimous consent.

Motion passes.

**6. Apply the changes in [P2017R1](#) (Conditionally borrowed ranges) to the C++ working paper.**

No discussion.

No objection to unanimous consent.

Motion passes.

**7. Apply the changes in [P1682R2](#) (std::to\_underlying for enumerations) to the C++ working paper.**

No discussion.

No objection to unanimous consent.

Motion passes.

**8. Apply the changes in [P0533R7](#) (constexpr for cmath and cstdlib) to the C++ working paper.**

Hubert Tong : I've put out a few emails last week to ask if certain considerations were discussed. I have not received confirmation either way. This is related to the treatment of error conditions in this paper.

Jonathan Wakely: LWG have touched on the lack of precision in which errors can prevent constant expression, the decision in LWG was to go ahead and put it on the straw polls page anyway. Hubert

has suggestions on how to make it more precise.

Davis Herring: we don't have the numerics chairs on the call.

Herb Sutter : we do have a LEWG and LWG chairs, and they have reviewed it. Bryce, do you have anything to add ?

Bryce Adelstein Lelbach : I don't believe this particular question has been discussed in LEWG.

Bjarne Stroustrup : Does anything block the improvements that you want ?

Hubert Tong : implementors need to understand how to implement this paper. The paper cites implementation experience, but it doesn't implement what is talked about. Implementors will not know what to implement and you will not get the improvements that you want. It will either not be implemented or you will get implementation divergence.

Bryce Adelstein Lelbach : LEWG approved this before C+20. The design of `is_constant_evaluated` hasn't settled yet, but we knew it was coming. Concerns were raised about implementability. We have more information now.

Jonathan Wakely : the implementors in LWG raised the concern too, but wanted this to go further and give feedback when they try to implement it. We need more implementors to look at it, and we may need to put in the draft to get that. We can take it out if we find it is not implementable. The paper has been there for 5 years and noone tried doing anything .

John Spicer : are there still objections in the room ?

There are objections in the room.

Herb Sutter clarifies the rules of voting

John Spicer clarifies how to vote.

In favour : 31

Opposed : 22

Neutral : 18

No consensus. Motion doesn't pass.

## 5.3 LEWG Polls

### Motion 1

Move to apply the changes in [P1919R3](#) (Expanding the Rights in SD-8) to Standing Document 8 (SD-8: Standard Library Compatibility).

Bryce Adelstein Lelbach : this is a paper that applies to a standing document, not to the standard or a TS. The SD8 was created by an LEWG motion. The chairs discussed this and we have decided that LEWG approval is all that is needed for this change and that the motion should come from LEWG.

No discussion.

No objection to unanimous consent.

Motion passes.



## **6. Closing activities**

### **6.1 Other business**

Jonathan Wakely reminds attendees to update their virtual meeting client.

### **6.2 PL2.16 motions, if any**

John Spicer reminds of voting rights for PL22.16

#### **1. TR 18015 National Review**

Barry : We have this TR which we planned on changing, but haven't gotten anywhere with it. We are going to re-affirm it. If nothing happens in the next 5 years, we will get rid of it.

Barry Hedquist moves.

Davis Herring seconds.

No objection to unanimous consent.

Motion approved.

## **7 Plans for the future (PL22.16)**

### **7.1 Next and following meetings**

Herb Sutter presents.

There is a question on the chat about the virtual meeting client poll feature. We tried it, and we couldn't make it work.

Thank you to hosts who have had to postpone. We are looking forward to seeing you in the future. When we have dates, we will inform everyone. We will have virtual meetings until then.

(virtual) 2021-06-07: Zoom virtual plenary meeting

(virtual) 2021-10-04: Zoom virtual plenary meeting

2022-02-07 to 12: Portland, OR, USA; Intel

2022-07: New York, NY, USA; Bloomberg

2022-11-07 to 12: Kona, HI, USA: Standard C++ Foundation

### **7.2 Mailings**

Note: These are the closest regular mailings and not special pre/post meeting mailings.

2021-06-15: Post-Summer

2021-09-15: Pre-Autumn

## 8. Adjournment (PL22.16 motion)

Daveed Vandevoorde moves

Davis Herring seconds

John Spicer adjourns the meeting at 09:22 AM N.Am. Pacific Time

## 9. Attendance

| Name                    | Organisation                   | National Body  |
|-------------------------|--------------------------------|----------------|
| Aaron Ballman           | Intel Corporation              | USA            |
| Abraham Berne           |                                |                |
| Alexandru Voicu         | AMD                            | USA            |
| Alisdair Meredith       | Bloomberg                      | United Kingdom |
| Andreas Weis            |                                | Germany        |
| Balint Joo              | Oak Ridge National Laboratory  | USA            |
| Barry Hedquist          | Perennial                      | USA            |
| Barry Revzin            | Jump Trading LLC               | USA            |
| Ben Saks                | Saks & Associates              | USA            |
| Bill Ash                | INCITS                         |                |
| Billy Baker             | FlightSafety International     | USA            |
| Bjarne Stroustrup       | Morgan Stanley                 | USA            |
| Botond Ballo            |                                | Canada         |
| Bronek Kozicki          |                                | United Kingdom |
| Bruno Cardoso Lopes     | Facebook                       | USA            |
| Bryan St. Amour         |                                | Canada         |
| Bryce Adelstein Leibach | NVIDIA Corporation             | USA            |
| Christian Trott         | Sandia National Laboratories   | USA            |
| Christof Meerwald       |                                | Austria        |
| Damien Lebrun-Grandie   | Oak Ridge National Laboratory  | USA            |
| Damon McDougall         | AMD                            | USA            |
| Dan Raviv               |                                |                |
| Daniela Engert          | GMH Prüftechnik GmbH           |                |
| Daveed Vandevoorde      | Edison Design Group            | USA            |
| David Olsen             | NVIDIA Corporation             | USA            |
| David Vitek             | GammaTech Inc                  | USA            |
| Davide Di Gennaro       |                                | Italy          |
| Davis Herring           | Los Alamos National Laboratory | USA            |
| Detlef Vollmann         |                                | Switzerland    |
| Dietmar Kühl            | Bloomberg                      | USA            |
| Dounia Khaldi           | Intel Corporation              | USA            |
| Drew Dormann            | Aquatic Group LLC              | USA            |
| Ellen Herrick           | Edison Design Group            |                |
| Eric Niebler            | Facebook                       | USA            |
| Erich Keane             | Intel Corporation              | USA            |
| Eugenio Bargiacchi      | Vrije Universiteit Brussel     |                |
| Fabio Fracassi          |                                | Germany        |
| Florian Sattler         |                                |                |

|                        |                                 |                |
|------------------------|---------------------------------|----------------|
| Frank Birbacher        |                                 | United Kingdom |
| Gabriel Dos Reis       | Microsoft France                | France         |
| Gašper Ažman           |                                 | United Kingdom |
| Graham Lopez           | NVidia Corporation              | USA            |
| Guy Davidson           |                                 | United Kingdom |
| Hal Finkel             | United States Dept of Energy    | USA            |
| Hana Dusíková          |                                 | Czech Republic |
| Hans Boehm             | Google                          | USA            |
| Harish Naik            | Aquatic Group LLC               | USA            |
| Herb Sutter            | Microsoft Corporation           | USA            |
| Howard Hinnant         | Ripple Labs                     | USA            |
| Hubert Tong            | IBM Corporation                 | Canada         |
| Inbal Levi             |                                 | Israel         |
| J. Daniel Garcia       | University Carlos III of Madrid | Spain          |
| Jason Merrill          | IBM Corporation                 | USA            |
| JC van Winkel          |                                 | Netherlands    |
| Jean-Paul Rigault      |                                 | France         |
| Jeff Garland           | Crystal Clear Software          |                |
| Jeff Snyder            |                                 | United Kingdom |
| Jens Maurer            | Edison Design Group             | USA            |
| Joe Sachs              | SAS Institute Inc               | USA            |
| John Lakos             | Bloomberg                       | USA            |
| John Spicer            | Edison Design Group             | USA            |
| Jonathan Caves         | Microsoft Corporation           | USA            |
| Jonathan Wakely        | IBM Corporation                 | United Kingdom |
| Joshua Berne           | Bloomberg                       |                |
| Juan Alday             | GreenWireSoft                   | USA            |
| Kelly Walker           | Stellar Science                 | USA            |
| Larry Lewis            | SAS Institute Inc               | USA            |
| Loïc Joly              |                                 | France         |
| Louis Dionne           | Apple                           | USA            |
| Lucian Radu Teodorescu |                                 | Romania        |
| Łukasz Wojakowski      |                                 | Poland         |
| Maged Michael          | Facebook                        | USA            |
| Marco Foco             |                                 | Italy          |
| Mark Hoemmen           | Stellar Science                 | USA            |
| Mark Zeren             | VMware Inc                      | USA            |
| Mateusz Pusz           | EPAM Systems Inc                | USA            |
| Matthew Butler         | Laurel Lye                      | USA            |
| Matthias Kretz         |                                 | Germany        |
| Michael Adams          |                                 | Canada         |
| Michael Davis          |                                 |                |
| Michael Hava           |                                 | Austria        |
| Michael Spencer        | Apple                           | USA            |
| Michael Spertus        | Amazon Corporate LLC            | USA            |
| Michael Wong           |                                 | Canada         |
| Michał Dominiak        | NVidia Corporation              | USA            |
| Mike Herrick           | Edison Design Group             | USA            |
| Mitsuhiro Kubota       |                                 | Japan          |
| Nathan Sidwell         |                                 | UK             |

|                        |                             |                |
|------------------------|-----------------------------|----------------|
| Nemanja Boric          | Amazon Corporate LLC        | USA            |
| Nevin Liber            | Argonne National Laboratory | USA            |
| Nicolai Josuttis       |                             | Germany        |
| Nina Ranns             |                             | United Kingdom |
| Pablo Halpern          | Halpern-Wight Inc           | USA            |
| Patrice Roy            |                             | Canada         |
| Paul Preney            |                             | Canada         |
| Peter Brett            |                             | United Kingdom |
| Peter Kulczycki        |                             | Austria        |
| Phil Ratzloff          | SAS Institute Inc           | USA            |
| Richard Corden         | Programming Research Ltd    | USA            |
| Rob Douglas            | Aquatic Group LLC           | USA            |
| Robert J. Simpson      | Qualcomm Inc                | USA            |
| Robert Leahy           |                             | Canada         |
| Roger Orr              |                             | United Kingdom |
| Samuel Ferreira Santos |                             |                |
| Scott Schurr           |                             |                |
| Stephen Upton          | Programming Research Ltd    | USA            |
| Steve Downey           | Bloomberg                   |                |
| Thomas Köppe           | Google                      |                |
| Tim Song               | Jump Trading LLC            | USA            |
| Tom Honermann          | Synopsys Inc                | USA            |
| Tony Van Eerd          |                             | Canada         |
| Vassil Vassilev        |                             | Bulgaria       |
| Ville Voutilainen      | The QT Company              | Finland        |
| Vittorio Romeo         |                             | Italy          |
| Wesley Maness          | Schonfeld Tools LLC         | USA            |
| William Miller         | Edison Design Group         | USA            |
| Wuping Xin             | KLD Engineering P.C.        | USA            |
| Zhihao Yuan            | SimpleRose Inc              |                |